SMALL SAMPLE TESTS

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 15

5-MINUTE REVISION

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1. QUICK REVISION · 1

- Small sample → often described as $n < 30$, but this is not a universal boundary.
- Small sample theory → study of sampling distributions for small samples.
- Important small-sample distributions → t, χ^2, and F.
- Student's t-test → used mainly for small samples when σ is unknown.
- Exact one-sample t-test → independent random observations from a normal population with unknown standard deviation.

2. QUICK REVISION · 2

- One-sample t-test → tests a population mean against a specified value.
- One-sample t degrees of freedom → $n - 1$.
- Two-sample t-test → compares two population means.
- Equal variances → use pooled variance.
- Equal-variance two-sample t degrees of freedom → $n_1 + n_2 - 2$.

3. QUICK REVISION · 3

- Unequal variances → use Welch's t-test.
- Welch's t-test uses separate sample variance terms.
- Welch's degrees of freedom are generally not an integer and are calculated using the Welch-Satterthwaite formula.
- Paired t-test → compares two related observations.
- Examples of paired data → before and after measurements on the same experimental unit.

4. QUICK REVISION · 4

- Paired t-test is performed on the differences within pairs.
- Paired t degrees of freedom → $n - 1$.
- Correlation test → tests whether the population correlation coefficient is zero.
- For the correlation test, $H_0 : \rho = 0$.
- Correlation test degrees of freedom → $n - 2$.

5. QUICK REVISION · 5

- Chi-square tests → based on the chi-square distribution.
- Chi-square tests for categorical counts differ from the parametric variance test for a normal population.
- Important chi-square tests → goodness of fit, independence of attributes, and population variance.
- Goodness-of-fit test → compares observed and expected frequencies.
- Goodness-of-fit null hypothesis → category probabilities follow the specified distribution.

6. QUICK REVISION · 6

- Goodness-of-fit test is right-tailed.
- Goodness-of-fit degrees of freedom → $n - 1$ when no parameter is estimated.
- Independence test → determines whether two attributes are independent.
- Contingency table → arrangement of frequencies for combinations of categorical attributes.
- Expected frequency → row total × column total ÷ grand total.

7. QUICK REVISION · 7

- $r \times s$ contingency table degrees of freedom → $(r - 1)(s - 1)$.
- 2×2 contingency table degrees of freedom → 1.
- Yates' correction → optional continuity adjustment for a 2×2 chi-square approximation; assess expected frequencies and consider an exact test for sparse data.
- Population variance test → tests a specified value of σ^2.
- Chi-square variance statistic → $\frac{(n-1)s^2}{\sigma_0^2}$.

8. QUICK REVISION · 8

- F-test → tests equality of two population variances.
- Placing the larger sample variance first is a convenience for a two-sided F-test with upper-tail area $\alpha/2$.
- With the larger variance in the numerator, $F \geq 1$; with fixed labels, F may be less than 1.
- F degrees of freedom → $n_1 - 1$ and $n_2 - 1$.
- In hypothesis testing, compare the calculated statistic with the appropriate table value.

9. QUICK REVISION · 9

- For a two-tailed t-test → reject H_0 when $|t|$ exceeds the critical value.
- For a right-tailed t-test → reject H_0 when t exceeds the critical value.
- For a left-tailed t-test → reject H_0 when t is less than the negative critical value.
- Goodness-of-fit and independence tests use the upper chi-square tail; a variance test may be one-sided or two-sided.
- For a two-sided F-test → use both $\alpha/2$ tails, or the equivalent larger-variance ratio with an upper $\alpha/2$ cutoff.

10. QUICK REVISION · 10

- William Sealy Gosset developed the t distribution and published under the name "Student".
- Student's t distribution has heavier tails than the normal distribution.

11. IMPORTANT FORMULAE · 1

Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

One-sample t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Degrees of freedom for one-sample t-test:

$$df = n - 1$$

12. IMPORTANT FORMULAE · 2

Pooled variance:

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

Two-sample t statistic for equal variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Degrees of freedom for equal-variance two-sample t-test:

$$df = n_1 + n_2 - 2$$

13. IMPORTANT FORMULAE · 3

Welch t statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch degrees of freedom:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2(n_1-1)} + \frac{s_2^4}{n_2^2(n_2-1)}}$$

Paired difference:

$$d_i = x_i - y_i$$

14. IMPORTANT FORMULAE · 4

Mean paired difference:

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$$

Variance of paired differences:

$$s^2 = \frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}$$

Paired t statistic:

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

15. IMPORTANT FORMULAE · 5

Degrees of freedom for paired t-test:

$$df = n - 1$$

Test statistic for significance of correlation coefficient:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Degrees of freedom for correlation test:

$$df = n - 2$$

16. IMPORTANT FORMULAE · 6

Chi-square goodness-of-fit statistic:

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Degrees of freedom for goodness-of-fit test:

$$df = n - 1 - m$$

Here n is the number of categories and m is the number of parameters estimated from the data; use $m = 0$ when all probabilities are specified.

17. IMPORTANT FORMULAE · 7

Expected frequency in an $r \times s$ contingency table:

$$E_{ij} = \frac{(A_i)(B_j)}{N}$$

Chi-square test of independence:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{s} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Degrees of freedom for an $r \times s$ contingency table:

$$df = (r-1)(s-1)$$

18. IMPORTANT FORMULAE · 8

Chi-square statistic for a 2×2 contingency table:

$$\chi^2 = \frac{n(ad-bc)^2}{C_1 C_2 R_1 R_2}$$

Yates' corrected chi-square statistic:

$$\chi^2 = \frac{n[\max(0, |ad-bc| - n/2)]^2}{C_1 C_2 R_1 R_2}$$

Chi-square test statistic for population variance:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

19. IMPORTANT FORMULAE · 9

Degrees of freedom for chi-square variance test:

$$df = n - 1$$

F statistic:

$$F = \frac{s_1^2}{s_2^2}$$

Degrees of freedom for F test:

$$df_1 = n_1 - 1, \qquad df_2 = n_2 - 1$$

20. WORKED EXAMPLE FROM THE CHAPTER

A plant breeder studied the relationship between plant height and grain yield in a sample of 18 rice genotypes and obtained a correlation coefficient of 0.6. Test whether the population correlation coefficient is significant.

Solution

Null hypothesis, H_0: ρ = 0 (population correlation coefficient is zero)

Alternative hypothesis, H_1: $\rho \neq 0$ (population correlation coefficient is not zero)

Sample correlation coefficient (r) = 0.6

Number of pairs (n) = 18

Under the bivariate normal model, the following t statistic is used to test zero correlation:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6\sqrt{18-2}}{\sqrt{1-0.36}} = \frac{0.6 \times 4}{\sqrt{0.64}} = \frac{2.4}{0.8} = 3.00$$

t_{cal} = 3.00. The table value of t for $n-2 = 16$ degrees of freedom at the 5% level of significance (two-tailed) is 2.120. Since the calculated value is greater than the table value, we reject the null hypothesis and conclude that there is a significant correlation between plant height and grain yield.

21. SELF-CHECK FROM THE CHAPTER

1. For a normal population with unknown variance, the standardised sample mean follows Student's _____ distribution. **[t]**
2. The degrees of freedom of a one-sample t-test are _____. **[$n-1$]**
3. The two-sample t-test assuming equal population variances uses a _____ variance estimate. **[Pooled]**
4. The two-sample t-test that does not assume equal population variances is called _____ t-test. **[Welch's]**